Operating Systems MCQ Question Bank

60 questions | Difficulty 0.5 to 10 (step 0.5, 3 questions per level) | 4 options each

Q1. [Difficulty: 0.5] Which of the following is an example of an operating system?

A) Linux
B) Photoshop
C) MySQL
D) Python

Answer: A

Q2. [Difficulty: 0.5] What does RAM stand for?

A) Read Always Memory
B) Random Access Memory
C) Random Active Memory
D) Rapid Access Module

Answer: B

Q3. [Difficulty: 0.5] Which part of the operating system interacts directly with the hardware?

A) Compiler
B) Text editor
C) Kernel
D) Shell

Answer: C

Q4. [Difficulty: 1] Which of the following is a CPU scheduling algorithm?

A) Round Robin
B) Binary Search
C) Bubble Sort
D) Merge Sort

Answer: A

Q5. [Difficulty: 1] What is a process?

A) A hardware device
B) A compiled binary file
C) A program in execution
D) A file stored on disk

Answer: C

Q6. [Difficulty: 1] Which of the following is non-volatile memory?

A) RAM
B) Hard disk
C) Cache
D) CPU registers

Answer: B

Q7. [Difficulty: 1.5] What is the main function of a shell?

A) Allocates physical memory to processes
B) Manages the CPU cache
C) Provides an interface between the user and the kernel
D) Compiles source code into machine code

Answer: C

Q8. [Difficulty: 1.5] Which of the following is NOT a standard process state?

A) Compiling
B) Waiting
C) Ready
D) Running

Answer: A

Q9. [Difficulty: 1.5] Which of the following is the fastest memory?

A) CPU registers
B) SSD
C) RAM
D) L2 cache

Answer: A

Q10. [Difficulty: 2] What is multiprogramming?

A) Running a program only after the previous one finishes
B) Compiling several programs together
C) Running a single program on several CPUs
D) Keeping several programs in memory at once to maximize CPU utilization

Answer: D

Q11. [Difficulty: 2] What is a deadlock?

A) Processes waiting indefinitely for resources held by one another
B) Main memory running out of space
C) The CPU being idle while waiting for I/O
D) A process terminating abnormally

Answer: A

Q12. [Difficulty: 2] Which scheduling algorithm is non-preemptive by nature?

A) Preemptive Priority
B) Round Robin
C) Shortest Remaining Time First
D) First Come First Served (FCFS)

Answer: D

Q13. [Difficulty: 2.5] What is the purpose of a system call?

A) To let user programs request services from the kernel
B) To compile a program
C) To speed up the CPU clock
D) To format a disk partition

Answer: A

Q14. [Difficulty: 2.5] Which statement about threads of the same process is correct?

A) Each thread has its own separate address space
B) They share the same address space and open files
C) Threads cannot share data with each other
D) Each thread has its own separate code section and heap

Answer: B

Q15. [Difficulty: 2.5] Which information is stored in a Process Control Block (PCB)?

A) The full source code of the program
B) The disk partition layout
C) The user's login password
D) Process state and program counter

Answer: D

Q16. [Difficulty: 3] Which scheduling algorithm can cause starvation of long processes?

A) Round Robin
B) First Come First Served
C) Lottery scheduling
D) Shortest Job First (SJF)

Answer: D

Q17. [Difficulty: 3] Paging eliminates which type of fragmentation?

A) Internal fragmentation
B) External fragmentation
C) Both internal and external fragmentation
D) Neither internal nor external fragmentation

Answer: B

Q18. [Difficulty: 3] What is the purpose of a mutex?

A) To allocate memory to a process
B) To schedule processes by priority
C) To increase the page size
D) To ensure only one thread is in a critical section at a time

Answer: D

Q19. [Difficulty: 3.5] In Round Robin scheduling, if the time quantum is extremely large, the algorithm behaves like:

A) SJF
B) LIFO
C) FCFS
D) Priority scheduling

Answer: C

Q20. [Difficulty: 3.5] What is a context switch?

A) Copying a file from one disk to another
B) Switching the CPU to a higher clock speed
C) Swapping the RAM modules
D) Saving the state of the running process and loading the state of another

Answer: D

Q21. [Difficulty: 3.5] Which of the following is a necessary condition for deadlock?

A) Spooling
B) Time slicing
C) Hold and wait
D) Preemption

Answer: C

Q22. [Difficulty: 4] What value does fork() return in the child process?

A) The child's PID
B) The parent's PID
C) -1
D) 0

Answer: D

Q23. [Difficulty: 4] Which context switch is generally faster?

A) Between threads of the same process
B) Between two different processes
C) Between two processes running on different CPUs
D) Both take exactly the same time

Answer: A

Q24. [Difficulty: 4] What is virtual memory?

A) Memory physically located on the CPU
B) A copy of RAM stored in the cache
C) A technique that lets a process run even if it is not entirely in physical memory
D) Extra RAM added through a USB device

Answer: C

Q25. [Difficulty: 4.5] What is a zombie process?

A) A process that is waiting for user input
B) A process whose parent has terminated
C) A terminated process whose entry stays in the process table until its parent reads its exit status
D) A process that uses 100% of the CPU forever

Answer: C

Q26. [Difficulty: 4.5] Which statement about a mutex and a semaphore is correct?

A) Semaphores can only be used between threads of one process
B) A mutex can hold a count greater than one
C) A mutex should be released only by the thread that acquired it
D) A binary semaphore can never be used for mutual exclusion

Answer: C

Q27. [Difficulty: 4.5] Belady's anomaly can occur with which page replacement algorithm?

A) FIFO
B) Optimal
C) LRU
D) Both LRU and Optimal

Answer: A

Q28. [Difficulty: 5] What is thrashing?

A) A cache miss on every instruction
B) Excessive paging that causes very low CPU utilization
C) The CPU running at maximum clock speed
D) The disk becoming completely full

Answer: B

Q29. [Difficulty: 5] What is an orphan process?

A) A process created by a system call
B) A running process whose parent has terminated; it is adopted by init/systemd
C) A process with no allocated memory
D) A process that has finished but is not yet cleaned up

Answer: B

Q30. [Difficulty: 5] What does a TLB (Translation Lookaside Buffer) do?

A) Holds the process control blocks
B) Caches recent page table entries to speed up address translation
C) Buffers data going to the disk
D) Stores the contents of recently used files

Answer: B

Q31. [Difficulty: 5.5] The Banker's algorithm is used for:

A) Deadlock prevention
B) Disk scheduling
C) Deadlock avoidance
D) CPU scheduling

Answer: C

Q32. [Difficulty: 5.5] What does copy-on-write do after fork()?

A) Swaps the parent's pages to disk
B) Parent and child share pages until one of them modifies a page
C) Copies all pages to the child immediately
D) Gives the child no memory until it calls exec

Answer: B

Q33. [Difficulty: 5.5] When is a spinlock preferred over a mutex?

A) When the critical section performs disk I/O
B) For very short critical sections on a multiprocessor system
C) When the thread should sleep while waiting
D) For long critical sections on a single-CPU system

Answer: B

Q34. [Difficulty: 6] What is the convoy effect in FCFS scheduling?

A) Short processes get stuck waiting behind one long CPU-bound process
B) Processes are executed in random order
C) The CPU is overloaded by too many interrupts
D) Many processes are waiting for the same page

Answer: A

Q35. [Difficulty: 6] Which of the following causes internal fragmentation?

A) Swapping a process to disk
B) Using a TLB
C) Allocating memory in fixed-size blocks or pages
D) Variable-size partitions leaving many small free holes

Answer: C

Q36. [Difficulty: 6] Which IPC mechanism is generally the fastest?

A) Pipes
B) Message queues
C) Sockets
D) Shared memory

Answer: D

Q37. [Difficulty: 6.5] What is priority inversion?

A) A high-priority process is blocked by a lower-priority process holding a needed resource
B) Low-priority processes always run before high-priority ones
C) Priority values wrap around after an overflow
D) Equal-priority processes run in reverse order

Answer: A

Q38. [Difficulty: 6.5] Which statement correctly distinguishes starvation from deadlock?

A) Starvation needs a circular wait; deadlock does not
B) They are the same thing
C) Deadlock affects only one process; starvation affects all
D) In starvation a process waits indefinitely while others keep running; in deadlock a set of processes wait on each other forever

Answer: D

Q39. [Difficulty: 6.5] A semaphore initialized to 0 is typically used for:

A) Letting any number of processes pass wait() immediately
B) Disabling interrupts
C) Mutual exclusion of a critical section
D) Making one process wait until another signals an event

Answer: D

Q40. [Difficulty: 7] Which technique is commonly used to solve priority inversion?

A) Priority inheritance
B) Demand paging
C) Spooling
D) Round Robin scheduling

Answer: A

Q41. [Difficulty: 7] What is a key drawback of cooperative (non-preemptive) multitasking?

A) Context switches are impossible
B) It requires special hardware timers
C) Processes cannot share memory
D) One process that never yields can hang the whole system

Answer: D

Q42. [Difficulty: 7] In Unix, what happens to hard links and soft (symbolic) links when the original file name is deleted?

A) Both links break
B) Both links keep working
C) The soft link breaks, while the hard link still gives access to the data
D) The hard link breaks, while the soft link still works

Answer: C

Q43. [Difficulty: 7.5] What does the exec() family of calls do?

A) Duplicates the address space of the process
B) Terminates the process immediately
C) Replaces the current process's memory image with a new program, keeping the same PID
D) Creates a new child process

Answer: C

Q44. [Difficulty: 7.5] Why do operating systems usually approximate LRU (e.g., clock / second-chance) instead of implementing it exactly?

A) LRU needs knowledge of future references
B) LRU suffers from Belady's anomaly
C) LRU cannot be used with paging
D) Exact LRU needs an update on every memory reference, which is too expensive

Answer: D

Q45. [Difficulty: 7.5] Why can the Optimal page replacement algorithm not be implemented in a real system?

A) It causes more page faults than FIFO
B) It requires knowledge of future page references
C) It suffers from Belady's anomaly
D) It needs too many frames

Answer: B

Q46. [Difficulty: 8] Why should a thread re-check its condition in a while loop (not an if) after waking from a condition variable wait?

A) The loop speeds up the context switch
B) The OS may delete the condition variable otherwise
C) The condition may no longer be true by the time it runs (spurious wakeups or another thread got there first)
D) wait() always returns immediately

Answer: C

Q47. [Difficulty: 8] What is the working set of a process?

A) All pages in its virtual address space
B) The pages it has referenced in a recent time window; keeping them in memory helps prevent thrashing
C) The pages shared with other processes
D) The pages swapped out to disk

Answer: B

Q48. [Difficulty: 8] What does the dirty (modified) bit in a page table entry indicate?

A) The page was modified and must be written to disk before it is replaced
B) The page contains corrupted data
C) The page is shared by several processes
D) The page is locked in memory

Answer: A

Q49. [Difficulty: 8.5] In a typical multilevel feedback queue scheduler, what happens to a process that uses its entire time quantum?

A) It is moved to a higher-priority queue
B) It stays at the same priority forever
C) It is terminated
D) It is moved to a lower-priority queue

Answer: D

Q50. [Difficulty: 8.5] In the readers-writers solution that gives readers priority, which group can starve?

A) Readers
B) Writers
C) Neither; starvation is impossible
D) Both groups equally

Answer: B

Q51. [Difficulty: 8.5] Why must the TLB be flushed on a context switch unless its entries are tagged with address-space identifiers (ASIDs)?

A) The TLB has too little capacity
B) The TLB stores only kernel pages
C) Physical addresses change during a context switch
D) Different processes use the same virtual addresses for different physical pages

Answer: D

Q52. [Difficulty: 9] What is a futex in Linux?

A) A system call used only to create threads
B) A file system type
C) A mechanism where locking is done in user space and the kernel is involved only under contention
D) A page replacement policy

Answer: C

Q53. [Difficulty: 9] What is the 'thundering herd' problem?

A) A burst of page faults after a context switch
B) A disk head repeatedly sweeping between innermost and outermost tracks
C) Many waiting processes are woken for one event, but only one can proceed, wasting CPU time
D) Too many processes created by fork()

Answer: C

Q54. [Difficulty: 9] Which data structure does the Linux Completely Fair Scheduler (CFS) use to pick the next task?

A) A red-black tree ordered by virtual runtime
B) A min-heap ordered by burst time
C) An array of 140 priority lists
D) A FIFO queue

Answer: A

Q55. [Difficulty: 9.5] What does the priority ceiling protocol guarantee?

A) It prevents deadlock and limits priority inversion to at most one critical section
B) It eliminates the need for locks
C) It prevents page faults
D) It guarantees fair scheduling for all processes

Answer: A

Q56. [Difficulty: 9.5] Why is LRU free from Belady's anomaly?

A) It uses the future reference string
B) It always replaces the page with the fewest references
C) It never causes a page fault
D) It is a stack algorithm: pages held with n frames are always a subset of those held with n+1 frames

Answer: D

Q57. [Difficulty: 9.5] How does vfork() differ from fork()?

A) vfork() never returns to the parent
B) The child shares the parent's address space and the parent is suspended until the child calls exec or exits
C) vfork() copies all memory before returning
D) vfork() creates a thread instead of a process

Answer: B

Q58. [Difficulty: 10] What is the ABA problem in lock-free programming?

A) A page being replaced and loaded again immediately
B) A compare-and-swap succeeds because a value went from A to B and back to A, although the state changed in between
C) Two threads acquiring locks A and B in opposite order
D) A process being scheduled twice in a row

Answer: B

Q59. [Difficulty: 10] What is the key idea of Read-Copy-Update (RCU)?

A) Every read copies the data into the CPU cache

B) Readers access data without locks while writers publish a modified copy and wait for old readers to finish

C) Writers update data in place while readers are blocked

D) Readers and writers must take the same mutex

Answer: B

Q60. [Difficulty: 10] What does a memory barrier (fence) do?

A) Prevents the compiler and CPU from reordering memory operations across it

B) Blocks access to kernel memory from user mode

C) Protects a memory region from being swapped out

D) Limits the amount of memory a process can allocate

Answer: A